7.6 Hyperbolic Rotations and Relativity

WORKSHEET

The question

The transformation T from $\begin{pmatrix} x \\ y \end{pmatrix}$ to $\begin{pmatrix} x' \\ y' \end{pmatrix}$ in two-dimensional space is given by

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cosh u & \sinh u \\ \sinh u & \cosh u \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix},$$

where u is a positive real constant. Show that the curve with equation $x^2 - y^2 = 1$ is transformed into itself. Find the equations of two straight lines through the origin which transform into themselves.

A line, not necessarily through the origin, which has gradient $\tanh v$ transforms under T into a line with gradient $\tanh v'$. Show that $v' = v + u$.

The lines ℓ_1 and ℓ_2 with gradients $\tanh v_1$ and $\tanh v_2$ transform under T into lines with gradients $\tanh v_1'$ and $\tanh v_2'$ respectively. Find the relation satisfied by v_1 and v_2 that is necessary and sufficient for ℓ_1 and ℓ_2 to intersect at the same angle as their transforms.

In the case when ℓ_1 and ℓ_2 meet at the origin, illustrate in a diagram the relation between ℓ_1, ℓ_2 and their transforms.

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself What identity do $\cosh u$ and $\sinh u$ satisfy, analogous to $\cos^2 u + \sin^2 u = 1$? And what are the addition formulas for $\cosh(u+v)$ and $\sinh(u+v)$?

STAGE 1 The hyperbola is invariant

Ask yourself Which lines through the origin are mapped to themselves? (Look for eigenvectors.)

STAGE 3 When are angles preserved?

Ask yourself The angle ψ between two lines with gradients m_1, m_2 satisfies $\tan\psi = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right|$. Simplify this when $m_i = \tanh v_i$. (*Hint:* put everything over $\cosh v_1 \cosh v_2$.)

Pause and try When is $\cosh(v_1 + v_2) = \cosh(v_1 + v_2 + 2u)$?

Try this next

1. **Composing boosts.** Show that $T(u)T(w) = T(u+w)$, where $T(u)$ is the matrix in the question. What is $T(u)^{-1}$?

2. **Time dilation.** A spaceship travels at 0.8 times the speed of light. Find its rapidity u and $\cosh u$. If 10 years pass on Earth, how much time passes on the ship? (Use $t_{\text{ship}} = t_{\text{Earth}} / \cosh u$.)

3. **Hyperbolic area.** Show that the area enclosed by the hyperbola $x^2 - y^2 = 1$, the x-axis and the line from the origin to $(\cosh v, \sinh v)$ is $\frac{v}{2}$, just as the corresponding area for a circle is $\frac{\theta}{2}$. Why must T preserve such areas?

4. **Squeeze.** Change coordinates to $p = x + y$, $q = x - y$. Show that T becomes $p \mapsto e^u p$, $q \mapsto e^{-u} q$, and that the hyperbola becomes $pq = 1$. Describe the transformation geometrically in these coordinates.

Further reading

- **Edwin Taylor & John Wheeler, *Spacetime Physics*** (W. H. Freeman). Special relativity

with rapidity and spacetime diagrams, for students with A-level mathematics.

- **Tevian Dray, *The Geometry of Special Relativity*** (CRC Press). Relativity as hyperbolic trigonometry, in exactly the spirit of this section.
- **Albert Einstein, *Relativity: The Special and the General Theory*** (various editions). Einstein's own popular account, still very readable.

The full guided solution, and what the question leads to at university, is in Section 7.6:
https://beyondstep.keithmcnulty.org/ch07/hyperbolic-rotations.html